

Precept 8

Goals

- Idea of first step analysis.
- Application of random walks.

Exercises

Exercise 1 (Marketing Buzz)

In order to kill off its direct competitor, the Fruity Yogurt store comes up with a new Froyo flavor: the Orange & Black Hoagie Frozen Yogurt (the secret ingredient is mayonnaise). To create a marketing buzz about this new flavor, the manager proposes the following deal on their website. Every 5 minutes, the price of a cup of this new Froyo flavor either gets doubled or halved (each outcome is equally likely) independently of the initial price. It is currently priced at \$128. You have decided on the following strategy: if the price goes under \$1 you will buy it; if it goes above \$15,000, you will stop following the sale and send an email to President Eisgruber about price gouging. Assuming that the sale lasts forever, is it more likely that you will end up buying Froyo or send an email to Pres. Eisgruber?

Exercise 2 (Lazy Random Walk)

Consider the drunk on Prospect Ave. example where the possible destination points are a and b , $a < b$. You are at some point $a \leq S_0 \leq b$ and each time you attempt to take a step, you either go one step right, one step left, or your attempt is not successful and you stay at the same point. The respective probabilities of these three cases are the same and equal to $1/3$. Assume each attempt is independent from the rest. The location after n attempts is S_n . Will you ever reach either a or b ? What is the probability that you arrive to b before a ? How many attempts does it take on average to arrive at one of the points?

Exercise 3 (The Great Escape) - time-permitting

Two blind mice, Mickey and Minnie, live on a one-dimensional, infinitely long island. They move independently according to simple random walks with parameters p_a and p_b . Unbeknownst to them, they were the subjects of pharmaceutical testing which rendered them blind.

As long as they are strictly less than 4 step sizes away from each other, they remain blind.

However, if they end up 4 or more step sizes apart from each other, they regain their sight and escape the island, returning to their owner Mr. Walt who lives on Disney Street.

An unfortunate side effect of the lab tests is that Mickey and Minnie explode if they bump into each other. They each take a step size of 0.5 at each time step—mice have short legs. They are currently two step sizes apart, but of course, they don't know this. What is the probability that they will return to Disney Street?